

Lecture 3.2: Testing of Random Number Generators

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Motivation and Intuition

Lecture 3.1 showed that a generator can achieve the maximal possible period yet still hide serious defects, as RANDU's collapse onto parallel planes demonstrated. Since the arithmetic structure of a generator does not, by itself, certify statistical quality, the only remaining option is to examine the generator's *output* directly and ask whether it is statistically distinguishable from a genuine Uniform(0, 1) sample. Two quite different things can go wrong. First, the individual values might not be spread evenly over $[0, 1]$ — this is a failure of **uniformity**. Second, even if each value looks fine in isolation, the *order* in which they appear might not be random — this is a failure of **independence**, and it is exactly the kind of defect that trapped RANDU. This lecture develops one representative test of each kind in full, together with brief descriptions of three further independence tests.

The Testing Problem

Given output $U_1, \dots, U_n$ of a generator, every test in this lecture addresses the same joint null hypothesis:

$$H_0 : U_1, \dots, U_n \stackrel{\text{iid}}{\sim} \text{Uniform}(0, 1).$$

Rejecting H_0 signals that the output can be statistically distinguished from an ideal uniform sample; because the alternative is left completely unspecified, no single test can reveal *which* aspect of H_0 has failed, which is exactly why a battery of complementary tests — some sensitive to shape, others to sequence — is used in practice rather than any one test alone.

Uniformity: The Chi-square Test

The most direct way to check whether values are spread evenly is to bin them and compare the observed counts to what evenness would predict. Partition $[0, 1]$ into k equal bins with observed counts O_i . Under H_0 , each of the n values falls into each bin with probability $1/k$, so $(O_1, \dots, O_k) \sim \text{Multinomial}(n; 1/k, \dots, 1/k)$, and the discrepancy between observed and expected counts is summarized by

$$\chi^2 = \sum_{i=1}^k \frac{(O_i - n/k)^2}{n/k}.$$

Theorem 2.1. Under H_0 , $\chi^2 \xrightarrow{d} \chi_{k-1}^2$ as $n \rightarrow \infty$ with k fixed.

Proof. By the multivariate central limit theorem applied to the multinomial counts, the vector of standardized deviations converges, $n^{-1/2}(O_i - n/k)_{i=1}^k \xrightarrow{d} N_k(0, \Sigma)$, with covariance matrix $\Sigma = \frac{1}{k}I_k - \frac{1}{k^2}\mathbf{1}\mathbf{1}'$. This matrix is *singular* — it annihilates the all-ones vector $\mathbf{1}$, since $\Sigma\mathbf{1} = \frac{1}{k}\mathbf{1} - \frac{1}{k^2} \cdot k \cdot \mathbf{1} = \mathbf{0}$ — and this singularity is not an accident: it reflects the fact that the counts O_i are not free to vary independently, since they must always sum to the fixed total n . Losing one “degree of freedom” to this constraint is exactly why the limiting chi-square distribution below has $k - 1$, not k , degrees of freedom.

Writing χ^2 as the quadratic form $n^{-1}(O - n\mathbf{p})'D^{-1}(O - n\mathbf{p})$, with $D = \text{diag}(\mathbf{p})$ and $\mathbf{p} = (1/k, \dots, 1/k)'$, one can check directly that the matrix $D^{-1}\Sigma$ is idempotent (that is, $(D^{-1}\Sigma)^2 = D^{-1}\Sigma$) with trace exactly $k - 1$. A standard fact about quadratic forms in a (possibly singular) multivariate Normal vector states that if $V \sim N_k(0, \Sigma)$ and $D^{-1}\Sigma$ is idempotent of rank r , then $V'D^{-1}V \sim \chi_r^2$; applying this fact to the limiting Normal vector above, with $r = k - 1$, gives the stated convergence. ■

Worked example. Suppose $n = 40$ generated values are placed into $k = 4$ equal bins with observed counts $O = (8, 12, 11, 9)$. The expected count per bin is $E = 40/4 = 10$, so

$$\chi^2 = \frac{(8-10)^2}{10} + \frac{(12-10)^2}{10} + \frac{(11-10)^2}{10} + \frac{(9-10)^2}{10} = \frac{4+4+1+1}{10} = 1.0.$$

With 3 degrees of freedom, the 5% critical value is 7.815; since 1.0 is far below this threshold, there is no evidence against uniformity — the observed counts are well within the spread expected by chance alone.

Uniformity: The Kolmogorov–Smirnov Test

A drawback of the chi-square test is that it requires an arbitrary choice of bins, and can be insensitive to smooth departures from uniformity that happen not to line up with the bin edges. The Kolmogorov–Smirnov test avoids binning entirely by comparing the *entire* empirical distribution function to the target directly. With empirical cdf $F_n(x) = n^{-1} \sum_i \mathbf{1}\{U_i \leq x\}$, the statistic

$$D_n = \sup_x |F_n(x) - x|$$

measures the single largest vertical gap between the step function F_n and the straight line $F(x) = x$. Because F_n only changes value at the observed order statistics $U_{(1)} \leq \dots \leq U_{(n)}$, the supremum must be attained at one of these jump points (approached either from the left or from the right), giving the computable form

$$D_n = \max_{1 \leq i \leq n} \max \left\{ \frac{i}{n} - U_{(i)}, U_{(i)} - \frac{i-1}{n} \right\}.$$

Kolmogorov (1933) showed that $\sqrt{n} D_n$ converges to a specific limiting distribution under H_0 ; the argument requires the theory of weak convergence of the empirical process to a Brownian bridge, a substantially deeper tool than anything used elsewhere in this lecture, and the proof is omitted.

Worked example. For the small ordered sample $u_{(1)}, \dots, u_{(5)} = 0.10, 0.35, 0.42, 0.68, 0.90$, compute both quantities at each i :

i	$u_{(i)}$	$i/5 - u_{(i)}$	$u_{(i)} - (i-1)/5$
1	0.10	0.10	0.10
2	0.35	0.05	0.15
3	0.42	0.18	0.02
4	0.68	0.12	0.08
5	0.90	0.10	0.10

The largest entry anywhere in the table is 0.18, attained at $i = 3$, so $D_5 = 0.18$: the third order statistic falls short of its “expected” position $3/5 = 0.6$ by this amount.

Independence: The Runs Test

Uniformity tests say nothing about order. A **run** is a maximal increasing or decreasing block in the sequence; intuitively, if the values are independent, the sequence should switch between rising and falling reasonably often, but not with perfect, suspicious regularity.

Theorem 2.2. Under H_0 (with a continuous distribution, so ties have probability zero), the number of runs R satisfies $E(R) = (2n - 1)/3$.

Proof. Call an interior position $i \in \{2, \dots, n - 1\}$ a *turning point* if U_i is a local maximum or local minimum relative to its two neighbours U_{i-1} and U_{i+1} . Because U_{i-1}, U_i, U_{i+1} are three continuous, exchangeable random variables, all $3! = 6$ orderings of their relative sizes are equally likely; exactly 4 of these 6 orderings place U_i either largest or smallest of the three (i.e. at an extreme), so $P(\text{turning point at } i) = 4/6 = 2/3$. Summing this probability over the $n - 2$ interior positions, the expected number of turning points is $E(T) = (n - 2) \cdot \frac{2}{3}$. Finally, every turning point marks the boundary between one run and the next, so the number of runs is always exactly one more than the number of turning points, $R = T + 1$. Combining,

$$E(R) = E(T) + 1 = \frac{2n - 4}{3} + 1 = \frac{2n - 1}{3}.$$

■

A more elaborate version of the same turning-point argument, tracking how nearby turning-point indicators are correlated with one another, gives $\text{Var}(R) = (16n - 29)/90$; the standardized statistic $Z = (R - E(R))/\sqrt{\text{Var}(R)}$ is then approximately $N(0, 1)$ for large n , allowing a formal test.

Worked example. For the ten values 0.51, 0.62, 0.40, 0.85, 0.33, 0.71, 0.29, 0.58, 0.44, 0.77, the signs of consecutive differences are +, −, +, −, +, −, +, −, +: the sequence switches direction at *every single step*, so every sign is its own run, giving $R = 9$. With $n = 10$, $E(R) = 19/3 = 6.33$ and $\text{Var}(R) = 131/90 = 1.456$, so

$$Z = \frac{9 - 6.33}{\sqrt{1.456}} = 2.21.$$

Since $|Z| = 2.21 > 1.96$, this is significant at the 5% level: the sequence alternates far more regularly than genuine randomness would produce (recall that under H_0 a turning point occurs with probability $2/3$ at each step, not with certainty), so this pattern is exactly the kind of excessive negative dependence the runs test is designed to catch.

Three Further Independence Tests

The gap, poker, and serial correlation tests below are classical, all traceable to Knuth's (1997) standard treatment of statistical tests for random sequences.

- **Gap test.** Fix an interval $[\alpha, \beta] \subset [0, 1]$ of probability $p = \beta - \alpha$. Under H_0 , the number of values skipped between one visit to $[\alpha, \beta]$ and the next is $\text{Geometric}(p)$ — this is simply the waiting-time distribution for a $\text{Bernoulli}(p)$ “success” of landing in the interval. A chi-square test comparing observed gap-length frequencies to this geometric law reveals clustering (too many short gaps) or repulsion (too many long gaps) relative to independence.

Numerical illustration. With $[\alpha, \beta] = [0, 0.5]$, so $p = 0.5$, suppose a sequence hits this interval at positions 3, 6, and 12. The first gap skips the 2 values at positions 4, 5, and the second gap skips the 5 values at positions 7, ..., 11. Under H_0 , $P(\text{gap} = 2) = (0.5)^2(0.5) = 0.125$ and $P(\text{gap} = 5) = (0.5)^5(0.5) = 0.015625$; a full test would compare many such observed gap lengths, across a long sequence, to these geometric probabilities using a chi-square statistic.

- **Poker test.** Group m consecutive decimal digits of each generated value into a “hand” and classify it by its repeated-digit pattern, exactly as one classifies a poker hand by rank. Under H_0 , elementary counting gives fixed probabilities for each pattern (e.g. for $m = 5$ digits: all different 0.3024, one pair

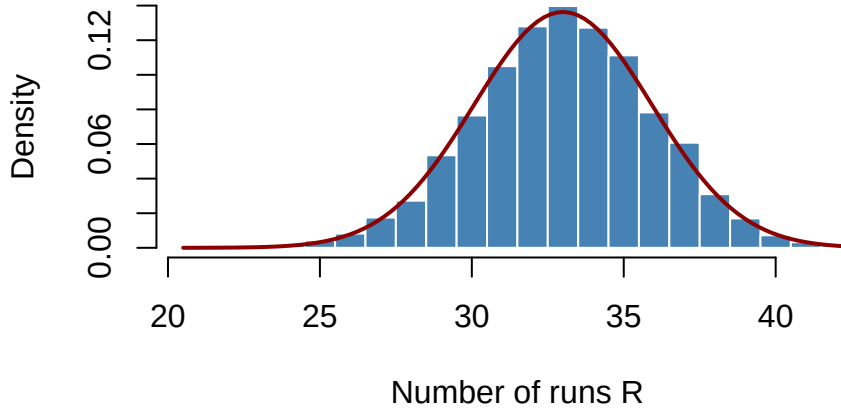


Figure 1: Simulated distribution of the number of runs ($n = 50$) against its Normal approximation from Theorem 2.2.

0.5040, two pairs 0.1080, three of a kind 0.0720, full house 0.0090, four of a kind 0.0045, five of a kind 0.0001), and a chi-square test compares the observed hand frequencies to these.

Numerical illustration. The value $u = 0.42517$ has digits 4, 2, 5, 1, 7 — all different, the most likely hand, with probability 0.3024 under H_0 . The value $u = 0.33218$ has digits 3, 3, 2, 1, 8 — the digit 3 repeats once, giving a “one pair” hand, with probability 0.5040 under H_0 ; a full poker test would classify many such hands this way and compare the resulting frequency table to these theoretical probabilities.

- **Serial correlation.** The lag- k sample autocorrelation r_k measures the linear association between values k steps apart; under H_0 it satisfies $r_k\sqrt{n-k} \xrightarrow{d} N(0,1)$, so values of r_k far from zero (relative to $1/\sqrt{n-k}$) indicate that successive (or k -apart) outputs are not behaving independently.

Numerical illustration. For the eight values 0.2, 0.5, 0.3, 0.8, 0.4, 0.6, 0.1, 0.9 (mean $\bar{u} = 0.475$), the lag-1 autocorrelation works out to $r_1 = -0.634$, and $Z = r_1\sqrt{8-1} = -1.68$. Since $|Z| = 1.68 < 1.96$, this particular short sequence does not give significant evidence of lag-1 dependence at the 5% level, even though $r_1 = -0.634$ looks large numerically — with only $n = 8$ values, sample correlations are highly variable under H_0 , and the formal Z -test, not the raw correlation, is what should be reported.

Each of these targets a different notion of “structure,” which is precisely why applying several of them together, as in the batteries used in practice, gives far more reliable protection than relying on any single test.

References

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- Knuth, D. E. (1997). *The art of computer programming, Volume 2: Seminumerical algorithms* (3rd ed.). Addison-Wesley.